

Cory A. Berger

NIH Postdoctoral Fellow

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Oakley Lab

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EDUCATION

Ph.D. in Biological Oceanography 2024

MIT-WHOI Joint Program in Oceanography, MA USA
Thesis: Exploring circadian rhythms, food intake, and their interactions in marine invertebrates
Committee: Drs. Ann Tarrant (advisor), Carolyn Tepolt, Greg Fournier, Casey Dunn

B.A. in Integrative Biology 2018

University of California, Berkeley, CA USA
Highest Honors in Integrative Biology, High Distinction in General Scholarship
Thesis advisor: Dr. Rosemary Gillespie

PROFESSIONAL APPOINTMENTS

NIH Postdoctoral Fellow 2026-present

University of California, Santa Barbara
Supervisors: Drs. Todd Oakley & Joel Rothman

Postdoctoral Researcher 2024-2026

University of California, Santa Barbara
Supervisor: Dr. Todd Oakley

PUBLICATIONS

10. **Berger, C. A.**, Stoilova, M. I., Varney, R. M., Abrams, S. C.[†], Miglietta, M., Cartwright, P., & Oakley, T. H. (2026). Adaptive molecular convergence is pervasive across deep time and largely decoupled from phenotypic convergence. *Proceedings of the National Academy of Sciences*, 123(31), e2616248123.
9. Picciani, N.^{*}, **Berger, C. A.**^{*}, Nielsen, S., Musser, J., Oel, A. P., Stoilova, M., Arendt, D., Garm, A., & Oakley, T.H. (2025). Comparative analysis of convergent jellyfish eyes reveals extensive differences in expression of vision-related genes. *Ecology and Evolution*, 15(7), e71784.
^{*}: equal contributions
8. Tarrant, A. M., & **Berger, C. A.** (2025). Cnidarian circadian clocks model how animals find predictability in a complex world. *Integrative and Comparative Biology*, icaf038.
7. **Berger, C. A.**^{*}, Steinberg, D. K., & Tarrant, A. M.^{*}. (2024). Nutritional condition drives spatial variation in physiology of Antarctic lipid-storing copepods. *Ecology and Evolution*, 14(9), e70210.
^{*}: equal contributions
6. **Berger, C. A.**, Steinberg, D. K., Copeman, L. A., & Tarrant, A. M. (2024). Comparative analysis of the molecular starvation response of Southern Ocean copepods. *Molecular Ecology*, e17371.
5. **Berger, C. A.**, & Tarrant, A. M. (2023). Sensory conflict disrupts circadian rhythms in the sea anemone *Nematostella vectensis*. *eLife*, 12, e81084.
4. **Berger, C. A.**, Ward, C. P., Karchner, S. I., Nelson, R. K., Reddy, C. M., Hahn, M. E., & Tarrant, A. M. (2022). *Nematostella vectensis* exhibits an enhanced molecular stress response upon co-exposure to highly weathered oil and surface UV radiation. *Marine Environmental Research*, 105569.

3. **Berger, C. A.**, Brewer, M. S., Kono, N., Nakamura, H., Arakawa, K., Kennedy, S. R., Wood, H. M., Adams, S. A., & Gillespie, R. G. (2021). Shifts in morphology, gene expression, and selection underlie web loss in Hawaiian *Tetragnatha* spiders. *BMC Ecology and Evolution*, 21(1), 1-17.
2. **Berger, C. A.**, Steinberg, D. K., Copley, N. J., & Tarrant, A. M. (2021). De novo transcriptome assembly of the Southern Ocean copepod *Rhincalanus gigas* sheds light on developmental changes in gene expression. *Marine Genomics*, 100835.
1. Arnold, I. J., **Berger, C.**, Chakrabarty, R. K., Moosmüller, H., Sharma, N., & Mazzoleni, C. (2014). The iron lung: A device for the continuous delivery of fine particulate matter. *Review of Scientific Instruments*, 85(2), 025105.

† denotes undergraduate mentee

PREPRINTS

Berger, C. A., & Tarrant, A. M. (2024). Feeding status modulates diel vertical migration of zooplankton via effects on circadian rhythms. *bioRxiv*, 2024.07.12.603318.

GRANTS, AWARDS, & FELLOWSHIPS

NIH F32 NRSA Postdoctoral Fellowship (2026-2029) “Determining the role of cell proliferation in mediating circadian effects on regeneration” Funding Agency: National Institute of General Medical Sciences (NIGMS)	\$237,708
Office of Research Early-Stage STEM Seed Grant (2024) “Establishing a molecular basis for the circadian regulation of regeneration” University of California, Santa Barbara	\$9,842
Fellowship in Graduate Studies (2022) “Metabolic regulation of diel vertical migration in <i>Acartia tonsa</i> (Copepoda)” The Crustacean Society	\$1000
Best student oral presentation, 1st runner up (2021) “Conserved molecular responses to starvation in two Southern Ocean copepods” The Crustacean Society	
Ocean Ventures Fund (2020) “The molecular basis of sensory integration in the circadian clock of <i>Nematostella vectensis</i> ” Woods Hole Oceanographic Institution	\$5,961
National Science Foundation Graduate Research Fellowship, Honorable Mention (2020)	
Summer Undergraduate Research Fellowship, UC Berkeley (2017) “Silk gene expression associated with loss of web-building in <i>Tetragnatha</i> spiders” Faculty Sponsor: Rosemary Gillespie	\$4,000

TEACHING EXPERIENCE

Co-instructor R workshop: Intro to Geospatial Raster and Vector Data Ocean Science Expedition Academy	Aug. 2026
Guest Lecturer Invertebrate Zoology (EEMB 116) University of California, Santa Barbara	Fall 2025
Instructor of Record Macroevolution (EEMB 102) University of California, Santa Barbara Positive teaching evaluations and comments from students available upon request	Spring 2025

Instructor certification, Software Carpentry	2024
Volunteer organization that teaches basic coding skills via hands-on workshops	
Teaching certificate, MIT Teaching + Learning Lab	Fall 2023
Completed interactive workshops on best practices in evidence-based, equitable pedagogy	
Covered course & lesson design, scaffolding & feedback, inclusive classroom environments	
Teaching Assistant	Fall 2021
Environmental Bioinformatics (EAPS 12.715)	
Massachusetts Institute of Technology	
Developed and led hands-on lab covering phylogenetic methods	
Software Carpentry Helper	
Workshop, plotting in R	Dec. 2021
Workshop, introductory R and git	Oct. 2021

CONFERENCE PRESENTATIONS

Cnidarian Model Systems Meeting (Poster)	Sep. 2026
Effects of circadian rhythms on whole-body regeneration in <i>Nematostella vectensis</i>	
Berger, C. A. , & Oakley, T. H.	
Evolution (Oral)	Jun. 2026
Adaptive molecular convergence is pervasive across deep time and largely decoupled from phenotypic convergence	
Berger, C. A. , Abrams, S., & Oakley, T. H.	
Society for Integrative and Comparative Biology (Oral)	Jan. 2026
Genomic analysis of Medusozoa reveals a decline in molecular convergence over deep timescales	
Berger, C. A. , Cartwright, P., Stoilova, M., Miglietta, M., Varney, R., & Oakley, T. H.	
Society for Integrative and Comparative Biology (Oral)	Jan. 2025
Phylogenomics provides insights into the repeated evolution of medusozoan eyes	
Berger, C. A. , Cartwright, P., Stoilova, M., Miglietta, M., Varney, R., & Oakley, T. H.	
Society for Integrative and Comparative Biology (Oral)	Jan. 2024
High-resolution tracking of zooplankton reveals metabolic control of diel vertical migration	
Berger, C. A. , & Tarrant, A. M.	
Cnidarian Model Systems Meeting (Oral)	Sep. 2022
How does <i>Nematostella</i> 's circadian clock behave under conflicting light and temperature cycles?	
Berger, C. A. , & Tarrant, A. M.	
ICES PICES Early Career Scientists Conference (Oral)	Jul. 2022
Integrating experimental and field-based gene expression measurements to identify metabolic biomarkers in Southern Ocean copepods	
Berger, C. A. , Steinberg, D. K., & Tarrant, A. M.	
US Antarctic Science Meeting (Oral)	Jul. 2021
Conserved and divergent gene expression changes in response to starvation in two Southern Ocean copepods	
Berger, C. A. , Steinberg, D. K., & Tarrant, A. M.	
Society for Integrative and Comparative Biology (Oral)	Jan. 2021
Conserved molecular responses to starvation in two Southern Ocean copepods	
Berger, C. A. , Steinberg, D. K., & Tarrant, A. M.	
Society for Integrative and Comparative Biology (Poster)	Jan. 2020
Can the cnidarian circadian clock entrain to temperature cycles?	
Berger, C. A. , & Tarrant, A. M.	
Pollutant Responses in Marine Organisms (Poster)	May 2019

- Transcriptomic profiling of a model cnidarian reveals enhanced stress response to weathered oil with exposure to ultraviolet radiation
Berger, C. A., Ward, C. P., Karchner, S. I., Reddy, C. M., Hahn, M. E., & Tarrant, A. M.
 Essig Undergraduate Research Symposium (Oral) May 2018
 Silk gene expression associated with web loss in *Tetragnatha* spiders
Berger, C. A., Adams, S., A., & Gillespie, R. G.
 Summer Undergraduate Research Fellowship Conference (Poster) Aug. 2017
 Silk gene expression associated with loss of web-building in *Tetragnatha* spiders
Berger, C. A., Adams, S., A., & Gillespie, R. G.

MENTORSHIP

University of California, Santa Barbara

- Graduate student mentor 2024-present
 Mentoring Ph.D. student C. McKinley on statistical and experimental methods of biological rhythms analysis
 Direct supervisor, undergraduate student researcher 2026-present
 Supervised research on regeneration and larval behavior in *Nematostella*, with a focus on image/video analysis
 Training in R, machine vision analyses of animal behavior, and RNA extractions
 Direct supervisor, undergraduate student researcher 2024-present
 Supervision of independent research project on the evolution of coloniality in Medusozoa
 Training in bash and R skills, character mapping & diversification rate analyses, phylogenetics
 Student has presented a poster at UCSB and is preparing to present at *Evolution* 2026; her work will result in co-authorship and first authorship on two upcoming papers
 Direct supervisor for undergraduate honors project (EEMB 116) 2024-2025
 Student conducted literature search on medusa stage evolution and learned R code for ancestral state reconstructions

Woods Hole Oceanographic Institution

- Mentored three students via Partnership in Education (PEP) program 2019, 2021, 2023
 PEP works with students from groups underrepresented in marine science
 Supervision of independent research projects; training in coding, benchwork, & presentation skills; advice on STEM careers & grad school
 Mentored two students via WHOI Summer Student Fellows program 2019, 2022
 Supervision of independent research projects; training in coding, benchwork, & presentation skills; advice on STEM careers & grad school
 Mentored four students via Sea Education Abroad Association program 2020
 Supervised students to prepare research plan for upcoming scientific cruise; advice on STEM careers & grad school

PRESS

- A toxic double whammy for sea anemones. *Oceanus* magazine. Article by Evan Lubofsky (2022)
<https://www.whoi.edu/oceanus/feature/oil-and-sun-can-be-a-tough-mix-for-sea-anemones/>
 Sea anemones with jet lag? Woods Hole Oceanographic Institution. Article by Evan Lubofsky (2019)
<https://www.whoi.edu/news-insights/content/sea-anemones-with-jet-lag/>

SERVICE & OUTREACH

Skype A Scientist	2026
Hosts virtual Q&A sessions about science careers with K-12 classrooms	
Cnidarian Model Systems Meeting Organizing Committee	2026
Writer and editor, OceanBites (https://oceanbites.org/)	2022-2024
Hosts blog posts summarizing primary literature for a general audience, receives thousands of visitors weekly	
I wrote 28 blog posts and edited many others	
Letters to a Pre-Scientist	2022-2023
Scientist pen-pal for students in schools from low-income communities	
High performance computing (HPC) advisory committee	2021-2023
Woods Hole Oceanographic Institution	
Committee for Diversity, Equity, and Inclusion: faculty recruitment subgroup	2020-2022
Woods Hole Oceanographic Institution	

Review Activities

Deutsche Forschungsgemeinschaft (German Research Foundation)
NSF Biological Oceanography Program
Arctic, Antarctic, and Alpine Research
Journal of Biological Rhythms
Journal of Experimental Biology
The Biological Bulletin

Professional Memberships

Society of Integrative and Comparative Biology
Society for the Study of Evolution

OTHER SKILLS & EXPERIENCE

Fluent (non-native) French speaker

NAUI open water diving certification

Sea-going cruise experience

R/V Neil Armstrong	Northeast US Continental Shelf
Chief Sci: Dr. Joel Llopiz	Nov. 12 – 14, 2018
SSV Corwith Cramer	Northeast US Continental Shelf
Chief Sci: Dr. Glen Gawarkiewicz	Jun. 27 – Jul. 6, 2018

Undergraduate Research Assistant

University of California, Berkeley

Feb. 2016 - May 2018
Advisor: Dr. Rosemary Gillespie

Lab Assistant, Zebrafish Facility

University of California, Berkeley

Apr. 2015 - May 2016
Supervisor: Melissa Boren

Research Assistant (Atmospheric Science)

Desert Research Institute, Reno

Sep. 2012 - Dec. 2013
Supervisors: Drs. Hans Moosmüller & Christopher Moore